

Algebra: Log Equations — Expanding & Condensing

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DIRECTIONS

Expand or condense each logarithmic expression using the properties of logs.

1 Expand the logarithm:

$$\log_2(8x^3)$$

Answer: _____

2 Condense into one logarithm:

$$3\log_2 x - \log_2 5$$

Answer: _____

3 Condense into one logarithm:

$$\frac{1}{2}\log x - \log 3$$

Answer: _____

4 Expand the logarithm:

$$\log\left(\frac{x^2}{y}\right)$$

Answer: _____

5 Expand the logarithm:

$$\ln(x^4y^2)$$

Answer: _____

6 Expand the logarithm:

$$\log\left(\frac{100}{x}\right)$$

Answer: _____

7 Expand the logarithm:

$$\log_3\left(\frac{x}{9}\right)$$

Answer: _____

8 Condense into one logarithm:

$$3\ln x - 2\ln y$$

Answer: _____

9 Condense into one logarithm:

$$\log 4 + 2\log x$$

Answer: _____

10 Condense into one logarithm:

$$2\ln x + \ln y$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES Expanding: split into separate logs. Condensing: reverse the rules. Domain must be $x > 0$.

1 Expand the logarithm:

$$= 3 + 3\log_2 x$$

$$\log_2 (8x^3)$$

2 Condense into one logarithm:

$$= \log_2 \left(\frac{x^3}{5} \right)$$

$$3\log_2 x - \log_2 5$$

3 Condense into one logarithm:

$$= \log \left(\frac{\sqrt{x}}{3} \right)$$

$$\frac{1}{2}\log x - \log 3$$

4 Expand the logarithm:

$$= 2\log x - \log y$$

$$\log \left(\frac{x^2}{y} \right)$$

5 Expand the logarithm:

$$= 4\ln x + 2\ln y$$

$$\ln(x^4 y^2)$$

6 Expand the logarithm:

$$= 2 - \log x$$

$$\log \left(\frac{100}{x} \right)$$

7 Expand the logarithm:

$$= \log_3 x - 2$$

$$\log_3 \left(\frac{x}{9} \right)$$

8 Condense into one logarithm:

$$= \ln \left(\frac{x^3}{y^2} \right)$$

$$3\ln x - 2\ln y$$

9 Condense into one logarithm:

$$= \log(4x^2)$$

$$\log 4 + 2\log x$$

10 Condense into one logarithm:

$$= \ln(x^2 y)$$

$$2\ln x + \ln y$$

